The Convergence Mirage: The Limits of Representation Similarity in From-Scratch Music Genre Classification

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Abstract

We benchmark a from-scratch architectural cross-section—SVM-RBF, Random Forest, XGBoost, 2D-CNN, BiLSTM, Transformer, Mamba-1, and Mamba-2—on the FMA Medium 16-genre task using a five-seed frozen evaluation protocol, and pair the comparison with a representation-similarity audit. SVM-RBF on a 540-dimensional classical feature matrix attains the highest macro-F1 (0.3994), and no pair of deep architectures is significant after Bonferroni correction. Penultimate-layer linear CKA among the five deep models is high (off-diagonal mean 0.6853), but a multi-init untrained-floor control places this only 0.0122 above a random-initialization baseline of 0.6731. The same architectures show large novelty gaps versus the classical feature matrix (+0.51 to +0.61), meaning that their mutual similarity is far larger than their similarity to hand-crafted DSP features. We treat this asymmetry as the central finding—a "convergence mirage"—and use it to reject H2 (learned representational convergence) as stated, qualify H1 as a limited Mamba-vs-Transformer small-data comparison, qualify H2a as a test of whether state-space inductive bias aligns with classical DSP features, and report H3 (per-architecture errors track feature-group CKA) as directionally supported. The paper's main limitation is that the FMA Medium label space caps macro-F1 through tiny rare-class supports, so the gap between classical and deep models may reflect label-space ceiling effects as much as model capability. A glossary of abbreviations and extended supporting tables appear in the Appendix.

Introduction

Genre classification is a canonical sequence task in music information retrieval: a track's waveform is mapped to one of a set of discrete genre labels through some intermediate representation. Beyond its value as a benchmark, genre tagging supports catalog search, recommendation, playlist organization, metadata repair, and archive-scale discovery when human annotations are sparse or inconsistent. More importantly for this paper, genre classification makes representation learning inspectable: if two architectures reach similar labels through different inductive biases, we can ask whether they learned the same musical abstractions or merely arrived at compatible decision rules. The space of intermediate representations now spans hand-crafted DSP features such as MFCCs and chroma (Tzanetakis and Cook 2002; Logan 2000; Bartsch and Wakefield 2005), convolutional spectrogram models (Choi et al. 2017), recurrent networks (Hochreiter and Schmidhuber 1997), attention-based encoders (Vaswani et al. 2017), and selective state-space models such as Mamba and Mamba-2 (Gu and Dao 2024; Dao and Gu 2024). Each family encodes a different hypothesis about what matters in audio—local spectrotemporal texture, sequential memory, global attention, or structured recurrence—so comparing them is also a test of what the architecture itself contributes before data scale and pretraining dominate the story. We include SSMs because they make a different claim about sequence modeling than attention: long-context processing through structured recurrence rather than all-pairs attention. Mamba-1 and Mamba-2 are both included because they represent adjacent SSM design points with similar parameter budgets but different state-space parameterizations.

Two kinds of question follow naturally. The first is performance: does any one of these inductive biases dominate on a controlled benchmark? The second is representational: do architecturally distinct deep models trained from scratch on the same data converge to a shared representation, and how does that representation relate to decades of hand-crafted MIR features? Recent work has begun to answer the representational question for vision (Raghu et al. 2021; Nguyen, Raghu, and Kornblith 2021), speech (Ollerenshaw, Jalal, and Hain 2022), and language (Hoang et al. 2025). For music genre classification trained from scratch, however, no published study evaluates the full spectrum—classical baselines through Mamba—under matched conditions with a representation-similarity audit and an explicit untrained-initialization control.

We close that gap on FMA Medium (Defferrard et al. 2017), holding dataset, splits, preprocessing, capacity band, optimizer family, and evaluation fixed across eight models (SVM-RBF, Random Forest, XGBoost, 2D-CNN, BiLSTM, Transformer, Mamba-1, Mamba-2) and five seeds, then probing penultimate-layer geometry with linear CKA (Kornblith et al. 2019) against trained, untrained, and within-architecture references. The empirical surface that emerges rejects the simplest version of the convergence story. Trained inter-architecture CKA is high in absolute terms but only marginally above an architectural-prior floor; deep models are simultaneously well separated from our 540-dimensional classical feature matrix; and classical baselines remain ahead on macro-F1.

We report this as a “convergence mirage”: much of the apparent representation-level agreement among deep models turns out to already be present at random initialization, so a heatmap of trained inter-architecture CKA cannot, on its own, be read as evidence of learned convergence. The same audit, however, surfaces a genuine and large novelty gap between every deep architecture and the classical feature matrix. By novelty gap, we mean the amount by which a deep architecture is more similar to other deep architectures than to the engineered DSP representation. It also finds a positive correlation between feature-group CKA profiles and confused-pair feature-space distances. The paper's contribution is therefore not a ”winner” among architectures, but a measurement-disciplined decomposition of which similarity signals are learned, which reflect inductive bias, and which are dominated by FMA's coarse 16-genre label space.

Research Questions and Hypotheses

We approached the empirical work around three research questions:

RQ1. How do classical baselines (SVM, RF, XGBoost) compare to from-scratch deep models (2D-CNN, BiLSTM, Transformer, Mamba-1, Mamba-2) on macro-F1 and related metrics under a controlled five-seed protocol on FMA Medium?

RQ2. How similar are the penultimate-layer representations of the five deep models—to one another, to the same architectures at random initialization, and to a 540-dimensional hand-crafted classical feature matrix?

RQ3. Do per-architecture feature-group CKA alignment profiles track the per-architecture pattern of which genre pairs are confused?

Here, a “confused” genre pair means that examples from one genre are repeatedly predicted as the other, producing high off-diagonal mass in the confusion matrix. We targeted this early because aggregate macro-F1 says which model wins, but recurring confused pairs reveal whether errors follow interpretable musical dimensions such as timbre, rhythm, or pitch-class structure.

The hypotheses below define guiding questions evaluated by the benchmarks and the representation-similarity audit.

H1 (Mamba vs. attention under matched conditions). On 30-second mel inputs, Mamba-2 will match or exceed Transformer macro-F1, testing whether selective state-space recurrence is competitive with attention in this small-data, from-scratch regime.

H2 (Representational convergence). Architecturally distinct deep models converge to a shared penultimate-layer representation.

H2a (SSM alignment with DSP features). Mamba penultimate representations are the most aligned with the classical DSP feature matrix, testing whether the state-space inductive bias recovers more hand-crafted MIR structure than the CNN, LSTM, or Transformer alternatives.

H3 (Interpretable error patterns). Per-architecture feature-group CKA alignment profiles correlate with the architecture's confused-pair feature-group distances.

The status of each hypothesis is explained across the Performance Results, Representation Analysis, and Error Analysis sections.

Related Work

Music genre classification. Hand-crafted DSP features—MFCCs, chroma, spectral contrast, rhythm descriptors—remain a strong baseline for music genre classification (Tzanetakis and Cook 2002; Logan 2000; Bartsch and Wakefield 2005; Müller and Ewert 2011; Jiang et al. 2002; Harte, Sandler, and Gasser 2006; Peeters 2004), especially when paired with kernel SVMs or gradient-boosted trees (Cortes and Vapnik 1995; Breiman 2001; Chen and Guestrin 2016). Modern deep approaches use 2D-CNNs and CRNNs over log-mel spectrograms (Choi et al. 2017), while sequence-modeling families—LSTMs, audio Transformers, and selective SSMs—have been imported from speech and language to audio (Vaswani et al. 2017; Gong, Chung, and Glass 2021; Gong et al. 2022; Gu and Dao 2024; Dao and Gu 2024; Yadav and Tan 2024; Shams et al. 2024). Benchmarks such as MARBLE (Benetos et al. 2023) compare music representation models, but pretraining is the central variable rather than from-scratch architectural inductive bias on a single corpus.

Sequence architectures and inductive bias. 2D-CNNs encode local time-frequency translation invariance (Choi et al. 2017); LSTMs sequentially update a finite hidden state (Hochreiter and Schmidhuber 1997); Transformers operate on full-sequence attention (Vaswani et al. 2017); structured state-space models including S4, Mamba, and Mamba-2 generalize linear recurrences with input-dependent gating, achieving linear-time sequence scaling (Gu, Goel, and Ré 2022; Gu and Dao 2024; Dao and Gu 2024). We therefore treat Mamba models as inductive-bias probes, not simply as newer competitors to tune for a leaderboard. From-scratch comparisons across the full spectrum—classical features through SSMs—on a single MIR benchmark with a representation-similarity audit have not, to our knowledge, been reported.

Representation similarity. Centered kernel alignment compares representations across models with different hidden dimensionalities under invariances that make it well-suited to cross-architecture studies (Kornblith et al. 2019). CKA has been used to study layer hierarchies in vision (Raghu et al. 2021; Nguyen, Raghu, and Kornblith 2021), speech (Ollerenshaw, Jalal, and Hain 2022), and language (Hoang et al. 2025). Two recurring caveats motivate our protocol: first, CKA between mean-pooled penultimate vectors confounds the sequence model with its pooling operator; we follow Raghu et al. (2021) in adopting this as a deliberate design choice and reporting alongside untrained and within-architecture references. Second, similarity scores must be compared against an architectural-prior floor. This control became central only after the trained CKA heatmap appeared highly convergent: random-initialization values can already be high, particularly across structurally related architectures, which can produce visually striking heatmaps without any learned alignment.

Audio data augmentation. SpecAugment (Park et al. 2019) masks contiguous time and frequency bands of a log-mel spectrogram and is now a default in audio classification pipelines. We use it as the sole augmentation in our deep training protocol and report a single-seed ablation with configuration caveats.

Method

The core pipeline is a controlled benchmark of eight models on a single FMA Medium split, instrumented for representation-similarity analysis. Implementation details and hyperparameters appear in the appendix.

Inputs

Deep models consume 30-second log1p-compressed mel power spectrograms of shape $(128, 1292)$, computed with a NumPy/SciPy STFT pipeline (window 2048, hop 512, 128 mel bins, mel filtering after power-spectrum computation, per-track z-score). Classical baselines consume a 540-dimensional feature matrix that combines FMA's precomputed 518-dimensional DSP feature table with a project-defined 22-dimensional rhythm vector derived with standard librosa tempo, onset, inter-beat-interval, and tempogram extractors (McFee et al. 2015). This added rhythm group fills a gap in the FMA feature table: FMA provides timbral, pitch, tonal, spectral, energy, and noisiness summaries, but not tempo, beat, onset, or metrical-structure descriptors. The same scaler, fit on the training split, is used for all classical baselines and as the CKA reference for deep models. The complete classical-feature table and the 8-group CKA taxonomy appear in the Appendix.

Models and Capacity Control

The five deep models live within a 2–5M parameter band so capacity does not trivially explain performance gaps: 2D-CNN (4.11M, 512-dim penultimate, 5 conv blocks); BiLSTM (2.44M, 512-dim, 2 layers, attention-pooled); Transformer (3.20M, 256-dim, 4 encoder blocks, [CLS] token, scaled dot-product attention); Mamba-1 (2.23M, 256-dim, 5 blocks); Mamba-2 (2.20M, 256-dim, 5 blocks). For each deep model, the post-pooling penultimate vector is used as the canonical representation for CKA, following the pooling-aware protocol of Raghu et al. (2021). SVM-RBF, Random Forest, and XGBoost have no parameter target. Per-architecture details are in the Architecture Summaries section of the Appendix. We intentionally avoid hybrid ensembles, pretrained encoders, and heavily optimized architecture variants because the comparison is about inductive bias under matched conditions, not maximum achievable genre-classification accuracy.

Training Protocol

Deep training uses AdamW (Loshchilov and Hutter 2019) with cosine annealing (Loshchilov and Hutter 2017), weighted cross-entropy with inverse-frequency class weights, batch size 32, mixed-precision autocast, and SpecAugment in training only. The 2D-CNN uses 40 epochs, patience 6, and no warmup; BiLSTM, Transformer, Mamba-1, and Mamba-2 use 50 epochs, patience 10, and five linear warmup epochs before cosine annealing. Per-model learning rate and dropout were chosen during a one-time tuning pass and frozen before final runs (Implementation Details in the Appendix). Best checkpoints are selected on validation macro-F1.

Frozen Evaluation Protocol

A single frozen evaluation set defines the 40 reported runs (5 deep architectures $\times$ 5 seeds plus 3 classical baselines $\times$ 5 seeds; SVM is deterministic and reported as a single point with seed-zero standard deviation by construction). Aggregation, paired tests, CKA, and figure rendering are all computed from this locked set.

For each deep pair we compute mean and standard deviation across seeds, run a paired test (paired t-test, falling back to Wilcoxon signed-rank if the Shapiro–Wilk $p < 0.05$), apply a Bonferroni correction at family-wise α=0.005 across the ten deep pairs, and report Cohen's d. SVM is excluded from significance tests and reported descriptively only because the RBF grid converges deterministically to the same setting across nominal seeds, leaving no comparable seed-level stochastic variation for paired testing.

CKA and Controls

Linear CKA between two centered representation matrices X, Y aligned on the same 1,772 test rows is $\mathrm{CKA}(X, Y) = \|Y^\top X\|_F^2 \,/\, (\|X^\top X\|_F \, \|Y^\top Y\|_F)$, equivalent to a normalized squared HSIC with linear kernels (Gretton et al. 2005; Kornblith et al. 2019). We report: (i) the trained inter-architecture off-diagonal mean of the 5×5 deep-model CKA matrix at seed 42; (ii) a single-init untrained floor (the same matrix with random initialization); (iii) a multi-init untrained floor averaged over five initialization seeds per architecture; and (iv) within-architecture cross-seed ceilings, which set an empirical upper bound for CKA at this scale. We define the per-architecture *novelty gap* as $\mathrm{gap}(a) = \overline{\mathrm{CKA}}_{\mathrm{inter}}(a) - \mathrm{CKA}(a, \mathrm{classical}_{540})$, the amount by which an architecture's average alignment with other deep models exceeds its alignment with the full classical feature matrix.

For RQ3 we additionally compute, per architecture, an 8-vector of CKA scores against each classical feature group, and Spearman ρ between this vector and the per-group genre-centroid distance restricted to the architecture's top-confused pairs.

Linear-Probe Transfer

To complement CKA with a supervised similarity probe we train class-balanced logistic-regression probes on each deep architecture's seed-42 penultimate representations, then evaluate off-diagonal transfer through a ridge linear bridge from the source representation space into the target space. Probes use an internal stratified 70/30 split of the test representations and are intended as a representation diagnostic, not a substitute for the frozen benchmark metrics. The full transfer matrix appears in Appendix .

Table 1: Headline performance on the FMA Medium test set (mean $\pm$ std across five seeds). SVM-RBF is deterministic and reports zero standard deviation by construction.

Model	Macro-F1	Accuracy	κ	MCC
SVM-RBF	**0.399 $\pm$ 0.000**	**0.722 $\pm$ 0.000**	**0.639**	**0.640**
XGBoost	0.373 $\pm$ 0.020	0.618 $\pm$ 0.032	0.532	0.537
Mamba-2	0.356 $\pm$ 0.021	0.592 $\pm$ 0.054	0.505	0.512
2D-CNN	0.356 $\pm$ 0.020	0.567 $\pm$ 0.048	0.476	0.487
Mamba-1	0.343 $\pm$ 0.047	0.579 $\pm$ 0.091	0.489	0.498
BiLSTM	0.335 $\pm$ 0.020	0.581 $\pm$ 0.042	0.486	0.491
Random Forest	0.323 $\pm$ 0.010	0.686 $\pm$ 0.001	0.561	0.578
Transformer	0.316 $\pm$ 0.024	0.550 $\pm$ 0.038	0.447	0.453

Experimental Setup

Dataset. FMA Medium (Defferrard et al. 2017): 17,000 30-second MP3 clips spanning 16 top-level genres. Twelve clips were unreadable under our decode protocol and excluded transparently, leaving **16,988** usable tracks. Splits follow the dataset-provided metadata and are artist-stratified with *zero* artist overlap between train, validation, and test partitions.

Splits. Benchmark-ready partition: 13,512 / 1,704 / 1,772 (train/validation/test). The 1,772-track test set is the canonical CKA alignment surface.

Class imbalance and metrics. The 16-genre distribution is severely imbalanced: on the benchmark-ready training split, Rock has 4,878 examples and Easy Listening has 13, a 375.2:1 ratio; over all usable tracks the ratio is 338.8:1. A majority-class baseline that always predicts Rock attains 0.3442 accuracy on the test split, while a uniform-random 16-way predictor attains 0.0625 expected accuracy. Macro-F1 is therefore the primary metric for class-imbalanced multiclass evaluation (Sokolova and Lapalme 2009); we additionally report accuracy, weighted F1, balanced accuracy, macro precision/recall, Cohen's κ, MCC, top-2/3 accuracy, expected calibration error (Guo et al. 2017), and Brier score (Brier 1950).

Baselines. SVM-RBF used grid search with balanced class weights; Random Forest and XGBoost used grid search (Chen and Guestrin 2016). We evaluated five seeds (42–46), with SVM repeated for table uniformity at zero variance because its grid converges deterministically on $(C{=}10, \gamma{=}0.001)$.

Compute. Experiments were run on a 12 GB consumer GPU (RTX 5070) and performant cloud GPUs (L40s, x4 RTX 5090 cluster). The same aggregation and evaluation procedure was used across the headline benchmark and robustness experiments.

Performance Results

Table 1 reports the headline 8-model performance summary on the frozen test set; full per-class metrics for all eight models appear in the Appendix.

Classical baselines lead. SVM-RBF on the 540-dimensional feature matrix attains the highest macro-F1 (0.3994) and the highest accuracy (0.7218), kappa (0.6389), and MCC (0.6399) of any model. XGBoost is second on macro-F1 (0.3727). The best deep model, Mamba-2 (0.3560), trails SVM-RBF by 0.044 macro-F1 absolute. Under our protocol, classical features remain hard to beat for 16-genre classification on FMA Medium.

Deep field is statistically indistinguishable. Mamba-2 and 2D-CNN are effectively tied at the top of the deep ranking (0.3560 and 0.3558), and Transformer is the weakest deep model (0.3161). Of the ten deep paired tests, *none* survive Bonferroni correction at $\alpha{=}0.005$. The three near-misses—2D-CNN vs. Transformer ($p{=}0.064$, $d{=}{+}1.14$), BiLSTM vs. Mamba-2 ($p{=}0.078$, $d{=}{-}1.05$), and Transformer vs. Mamba-2 ($p{=}0.085$, $d{=}{-}1.02$)—show large effect sizes that our five-seed budget cannot promote to significance under conservative correction. **H1** is therefore only directionally supported: Mamba-2 is 0.040 macro-F1 above Transformer with $|d|{>}1$, but the result is best read as evidence that Mamba-2 is competitive with attention under this controlled small-data setting, not as a general statement that Mamba architectures dominate Transformers.

Mamba-1 variability and Random Forest's accuracy gap. Mamba-1 has the largest seed-to-seed standard deviation among deep models (0.0471, more than twice the next), an instability we report; we did not drop or refit any frozen seed. Random Forest illustrates a separate caution: its accuracy is 0.6860, third-highest overall, while its macro-F1 is only 0.3226. Accuracy thus systematically misranks the imbalance behaviour of these models, which is why we lead with macro-F1.

Calibration. Table 2 summarizes calibration for the five deep models. 2D-CNN has the best ECE (0.0435), Mamba-2 has the best Brier score (0.5666), and Transformer has the worst Brier (0.6109) and the highest mean entropy (1.366), consistent with its weakest macro-F1. Top-2 and top-3 accuracies cluster between 0.72–0.75 and 0.80–0.82 respectively across deep models, indicating that discriminative information exists for the long tail but is suppressed at the top-1 threshold.

Representation Analysis

This is the central CKA-based audit. The headline numbers are summarized in Table 3; Figures 1, 2, and 3 display the inter-architecture matrix, the novelty gap, and the 8-group profile respectively.

Trained Inter-Architecture Similarity

Trained inter-architecture linear CKA averages 0.6853 over the ten deep-model pairs—high in absolute terms and consistent with prior cross-architecture similarity findings in vision (Raghu et al. 2021; Nguyen, Raghu, and Kornblith 2021) and language (Hoang et al. 2025). The corresponding multi-init untrained floor, computed by re-running the same penultimate extraction on five randomly initialized copies of each architecture, is 0.6731. Trained CKA exceeds the

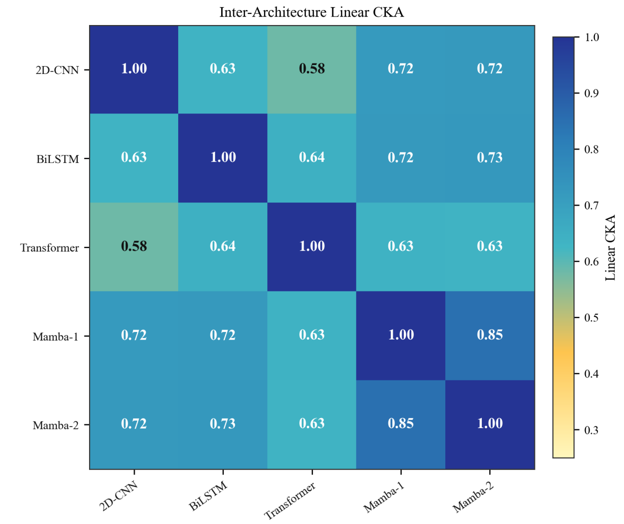

Figure 1: Trained inter-architecture linear CKA at the penultimate layer over 1,772 test samples (seed-42 representatives; off-diagonal mean 0.685). Visually striking, but only 0.012 above the multi-init untrained floor of 0.673 (Representation Analysis).

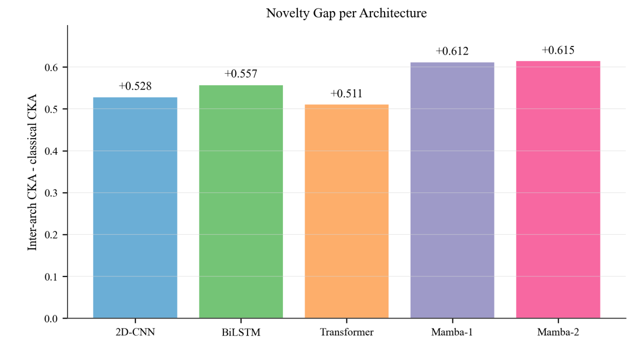

Figure 2: Novelty gap per architecture: the difference between the architecture's mean inter-architecture CKA and its CKA against the 540-d classical feature matrix. All five deep models share a representational subspace not captured by classical DSP features.

floor by only 0.0122; the single-init floor (0.6643) yields a similarly small lift of 0.0210. Within-architecture cross-seed CKA averages 0.7990 across the five archs (2D-CNN 0.877, BiLSTM 0.830, Mamba-2 0.790, Mamba-1 0.750, Transformer 0.748), the canonical sanity-check ordering inter-arch $<$ within-arch.

The floor is not uniform: pairs that include 2D-CNN sit at 0.30–0.34, while sequence-style pairs (BiLSTM, Transformer, Mamba-1, Mamba-2 against each other) already reach 0.84–0.94 at random initialization. The 5-init pair statistics (e.g. BiLSTM–Mamba-1 0.938 ± 0.003, Transformer–Mamba-1 0.934 ± 0.011, Mamba-1–Mamba-2 0.917 ± 0.015, 2D-CNN–BiLSTM 0.305 ± 0.026) show that the apparently uniform 0.685 trained mean is the average of two regimes: a low CNN-vs-sequence regime and a high sequence-vs-sequence regime. Reading the trained heatmap (Figure 1) without this control is what we mean by the convergence mirage—most of the inter-architecture similarity is structural and pre-training, not learned. This late-stage control changed the interpretation of the study: what first looked like strong learned convergence became a much smaller training effect sitting on top of a high architectural-prior floor. **H2** (representational convergence to a shared learned subspace) is therefore not supported as stated; we instead report high inter-architecture CKA dominated by an architectural prior.

The Novelty Gap

Despite the small trained-vs-floor lift, deep representations are far from the hand-crafted classical feature matrix. Per-architecture linear CKA against the 540-dimensional scaled classical matrix is uniformly low (0.111–0.134) while the same architectures' inter-architecture mean is 0.62–0.73, producing per-architecture novelty gaps of +0.51 to +0.61 (Figure 2). The five deep models therefore agree on *some* representational structure that the classical feature matrix does not capture, even if how much of that agreement is learned versus prior is small.

Mamba-1 (+0.612) and Mamba-2 (+0.615) have the largest novelty gaps, exceeding Transformer (+0.511). The driver is inter-architecture CKA, not classical alignment: raw CKA-to-classical is essentially flat (0.111–0.134), and *2D-CNN* has the highest raw classical CKA at 0.1339, not Mamba. **H2a** as originally written—"Mamba is the most classical-feature-aligned architecture"—is therefore not supported. The salvageable, evidence-faithful version of H2a is that Mamba representations are most aligned with the deep-model cluster.

Feature-Group Profiles

Decomposing CKA against the eight classical feature groups (Figure 3) shows that timbre (the 140-dimensional MFCC group) and rhythm have the strongest deep-vs-classical alignment, while pitch-class (chroma) is the weakest. Across architectures, timbre CKA ranges 0.36–0.39 and pitch-class CKA ranges 0.07–0.09. 2D-CNN's rhythm CKA (0.395) is noticeably higher than every other architecture's, mirroring its strong inductive bias for local time-frequency structure. We treat group-level interpretation as correlational, as mechanistic claims ("2D-CNN learns rhythm") would require feature-attribution or feature-group dropout experiments not present in this research.

Cross-Architecture Linear Probe Transfer

A complementary supervised diagnostic agrees with the novelty-gap reading. Class-balanced logistic-regression probes trained on each architecture's seed-42 representations attain a mean self-probe macro-F1 of 0.5352; mean off-diagonal transfer through a ridge linear bridge attains 0.4926, a 92.0% retention. Best off-diagonal transfer is BiLSTM→2D-CNN (0.5617); worst is 2D-CNN→Transformer (0.4058). The retention level is consis-

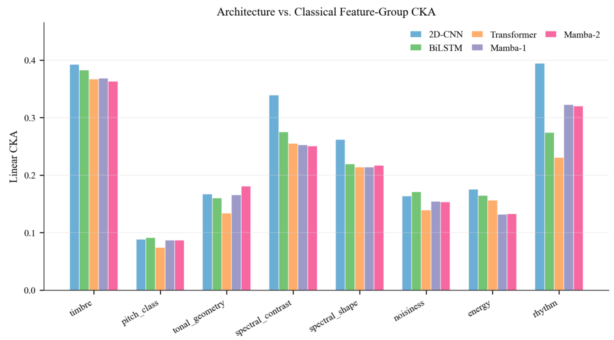

Figure 3: Feature-group CKA profiles: per-architecture linear CKA against each of the eight classical feature groups. Timbre and rhythm dominate; pitch-class is uniformly weak.

tent with substantial shared class-discriminative geometry across architectures, although it is seed-42 only and uses an internal 70/30 split of the test representations rather than the frozen test surface, so it serves as supporting representation evidence rather than as a benchmark metric. Appendix reports the full transfer matrix.

Error Analysis

Per-class behaviour is dominated by support. Old-Time / Historic is the easiest class across all eight models (per-class F1 0.86–0.98), with Random Forest and SVM-RBF reaching 0.97–0.98. Easy Listening (test support 6) and International (test support 2) are zero-F1 across every model, including all five seeds of every deep architecture and all three classical baselines. Pop is near-zero (range 0.00–0.05) and Blues is zero for many models but not universally—2D-CNN attains a Blues F1 of 0.0901 and Transformer 0.0200. Inverse-frequency class weighting was insufficient for the long tail; the appropriate framing is a label-space limitation, not a model-family failure.

Recurring confusions are partly real, partly support-driven. Table 4 lists, for each deep architecture, its top-3 confused pairs by normalized off-diagonal mass; the full top-5 table per architecture, with per-feature-group genre-centroid distances, is in the Appendix. Rock $\leftrightarrow$ Soul-RnB appears in the top-five of *all five* deep architectures, the only pair to do so. Other recurring pairs include Folk $\leftrightarrow$ Instrumental, Classical $\leftrightarrow$ Easy Listening, and International-anchored pairs that are largely driven by the tiny International test support (n=2): Transformer's International $\leftrightarrow$ Jazz mass of 0.815 is the largest confusion mass observed in any model, but with only two International test samples, any redirection produces extreme normalized values.

H3: feature-group CKA tracks confusion structure, directionally. For each architecture we computed Spearman ρ between (a) its 8-vector of feature-group CKA scores and (b) its mean per-group genre-centroid distance over its top-5 confused pairs. All five ρ are positive: 2D-CNN +0.62, BiLSTM +0.76, Transformer +0.67, Mamba-1 +0.71, Mamba-2 +0.71. Three of five (BiLSTM, Mamba-1, Mamba-2) reach uncorrected $p < 0.05$; with only eight feature groups per architecture, none survive a conservative correction. We therefore report **H3** as directionally supported—not because any single small-n test is decisive, but because all five architectures share the same sign, suggesting that feature-group alignment and error geometry move together across otherwise different inductive biases.

Robustness and Ablations

15-second crop. Center-cropping the 30-second mel input to 15 seconds reduces macro-F1 by 0.027–0.047 for 2D-CNN, BiLSTM, and Mamba-1, but leaves Transformer (-0.001) and Mamba-2 (-0.005) essentially unchanged. Because the crop ablation combines two preprocessing sources, the comparison is interpretive; the cross-architecture ordering is the load-bearing observation rather than the absolute deltas. Full table in the Appendix.

SpecAugment. At seed 42, removing SpecAugment costs 2D-CNN 0.111 macro-F1 and Transformer 0.050; Mamba-2 changes by only -0.010; BiLSTM is paradoxically 0.015 better without SpecAugment. The Mamba-1 no-SpecAugment run also changed learning rate (0.001 to 0.0002) and early-stopping patience (10 to 8), so its -0.010 delta is not an augmentation-only estimate. We treat the BiLSTM number and the no-Spec runs as single-seed sensitivity checks rather than general architecture claims. The qualitative representation story persists, while augmentation sensitivity varies sharply for the architectures whose runs are cleanly matched.

60-second context. A long-sequence pilot on 60-second inputs (Mamba-2 at seeds 42–46, Transformer at seeds 42–44) does not uniformly improve sequence models. Transformer macro-F1 rises from 0.3161 (30s) to 0.3281 (60s, +0.012); Mamba-2 falls from 0.3560 (30s) to 0.3306 (60s, -0.025). Preprocessing differs from the 30-second baseline and seed counts are unbalanced, so we report this as a robustness check rather than a definitive context-length ranking.

Discussion and Limitations

The mirage and its scope. The central interpretive claim of this paper is that a high inter-architecture CKA score among from-scratch deep audio classifiers is, on its own, insufficient evidence of learned representational convergence. The 0.6853 trained mean carries a small 0.0122 lift above a multi-init untrained floor that is itself bimodal in architecture-family terms. The mirage is asymmetric, however: the same representations are well separated from the classical feature matrix (novelty gaps +0.51 to +0.61), and a supervised probe transfer retains 92% of self-probe macro-F1 across architectures. The defensible reading is that deep models share a representational subspace that classical DSP features do not capture, but most of *how much* they share is structural inductive bias, not learning.

Label-space ceiling. On FMA Medium, the 16-class taxonomy itself caps macro-F1. Easy Listening (test support 6) and International (test support 2) are zero-F1 across every model; Pop and Blues are near-zero. This cap is one mechanism by which classical baselines remain ahead: SVM-RBF's strong rare-class behaviour suffices to dominate on macro-F1 even though deep models learn richer-but-correlated representations. A different label scheme, or a different evaluation surface that handled tiny-support classes, could change the macro-F1 ranking even with identical models.

From-scratch data regime. Some deep-model degradation is expected under this setup. Transformers are often data-hungry, and all five deep architectures are trained from scratch on a modest, imbalanced benchmark rather than initialized from a large audio-pretraining corpus; recent music-representation work likewise treats limited labeled data as a central constraint (Plachouras, Benetos, and Pauwels 2025). We deliberately do not add hybrid CNN–Transformer/SSM systems, pretrained encoders, or architecture-specific tuning tricks, because doing so would blur the inductive-bias comparison that motivates the paper.

Statistical power. Five seeds are not enough to promote large-effect-size deep-vs-deep differences (Cohen's $|d|>1$ for three pairs) past Bonferroni correction at $\alpha=0.005$. We chose to report effect sizes alongside raw p-values rather than expand the seed budget, but a larger benchmark would clarify whether 2D-CNN $\approx$ Mamba-2 is a genuine tie or simply an under-powered comparison.

Pooling confound. CKA is computed on post-pooling penultimate vectors, which means "representation similarity" includes the effect of each architecture's pooling operator (mean for Mamba, attention-pool for BiLSTM, [CLS] for Transformer, adaptive average for 2D-CNN). Following Raghu et al. (2021), this is deliberate: downstream classifiers see post-pooling features. Disentangling pre-pooling sequence representations is left to future work.

Future feature-ablation test. The most direct missing analysis is an SVM feature-group dropout study: retrain the winning SVM-RBF baseline while removing each classical feature group in turn, then measure the macro-F1 drop. That experiment was not run here, but it would test which handcrafted feature families explain the classical baseline's advantage more directly than CKA alignment alone.

Robustness scope. The headline benchmark and representation analysis use the same locked 40-run evaluation set. The robustness checks are reported separately because they vary context length, crop length, or augmentation status rather than the headline model family comparison.

Conclusion

Do these architectures converge through learning? Barely, and not in the way the hypothesis predicted. After training, the five deep models reach an off-diagonal CKA mean of 0.685, which sounds like convergence until the untrained control is included: random networks with the same architectures already reach 0.673. Training moves the cross-architecture similarity by about one CKA point, so the strongest version of learned convergence is not supported.

The stronger result is the one next to that negative finding. The same deep models have CKA only 0.111–0.134 against the 540-dimensional classical feature matrix, producing novelty gaps of +0.51 to +0.61 across every architecture. In plain terms, the models start similar and end similar, but the shared structure they express is not the structure captured by MFCCs, chroma, spectral summaries, and our added rhythm descriptors. Deep models share a representation of music that hand-crafted DSP features miss, even though that shared representation does not beat the SVM-RBF baseline on macro-F1 in this label regime.

The final takeaway is a measurement-discipline correction for representation studies: a trained similarity heatmap is not enough. Without an architectural-prior floor, 0.685 CKA looks like a discovery; with the floor, it becomes a convergence mirage with a small learned component. For MIR, this makes representation audits more useful, not less: future work should report untrained floors, same-architecture ceilings, kernel checks, supervised transfer probes, and feature-group ablations before treating cross-model similarity as evidence of learned convergence.

What, then, is the shared deep-model space? The cautious answer is that it is not simply MFCCs, chroma, spectral summaries, or our added rhythm descriptors in another coordinate system. A more speculative answer is that the deep models are compressing a mid-level musical "texture" space: local timbral envelopes, onset density, stationarity, coarse temporal shape, and label-correlated production cues that are distributed across handcrafted feature groups rather than isolated inside any one of them. This interpretation is consistent with auditory texture work showing that summary statistics over cochlear envelopes and modulation structure can carry perceptually meaningful sound identity (McDermott and Simoncelli 2011), and with MIR arguments that deep feature learning can move beyond fixed descriptors toward task-shaped audio representations (Humphrey, Bello, and LeCun 2013). That would explain why the space transfers across architectures, remains far from classical DSP summaries, and still does not automatically win macro-F1 under a rare-class ceiling. The defensible story is therefore narrower but more interesting: from-scratch deep audio classifiers do not convincingly converge to a shared learned subspace, but they do consistently occupy a deep-model representation space that classical audio features fail to explain.

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Appendix

This Appendix documents (A) abbreviations, (B) implementation details, (C) architecture summaries, (D) the complete classical-feature table and 8-group CKA taxonomy, (E) full per-class metrics for all eight models, (F) the complete top-5 confusion table with per-group genre-centroid distances, (G) full multi-init untrained-floor and trained-vs-untrained tables, (H) the per-genre inter-architecture CKA breakdown, (I) the full pairwise linear-probe transfer matrix, (J) full robustness and ablation tables, (K) extended visual diagnostics, and (L) reproducibility notes.

Abbreviations

Abbrev.	Meaning
MIR	Music information retrieval
FMA	Free Music Archive (Defferrard et al. 2017)
DSP	Digital signal processing
MFCC	Mel-frequency cepstral coefficient
SSM	State-space model
CKA	Centered kernel alignment (Kornblith et al. 2019)
HSIC	Hilbert–Schmidt independence criterion
ECE	Expected calibration error
MCC	Matthews correlation coefficient
AUC	Area under the ROC curve
RBF	Radial basis function (kernel)
SVM	Support vector machine
RF	Random forest
STFT	Short-time Fourier transform
[CLS]	Classification token (Transformer)

Implementation Details

Optimization. AdamW (weight decay 10^{-4}) with cosine annealing; BiLSTM/Transformer/Mamba-1/Mamba-2 use five linear warmup epochs and 2D-CNN uses no warmup. Weighted cross-entropy uses inverse-frequency class weights computed from the training split. Batch size 32, 4 dataloader workers, mixed-precision autocast, and GPU memory fraction 0.75 are shared. 2D-CNN uses max 40 epochs and patience 6; BiLSTM/Transformer/Mamba-1/Mamba-2 use max 50 epochs and patience 10. SpecAugment (training only): two frequency masks $\leq$ 20 bins and two time masks ≤ 50 frames.

Per-model learning rate / dropout. 2D-CNN: lr 3×10^{-4}, dropout 0.0. BiLSTM: 5×10^{-4}, 0.3. Transformer: 5×10^{-4}, 0.1. Mamba-1: 1×10^{-3}, 0.1. Mamba-2: 1×10^{-3}, 0.1. SVM-RBF: grid converged on $(C{=}10, \gamma{=}0.001)$ each seed; class-balanced loss. Random Forest and XGBoost: grid-tuned, class-weighted.

Mamba-1 stability note. For the crop ablation, Mamba-1 used a lower learning rate than the headline run to maintain stable optimization under the shorter input setting: 2×10^{-4} rather than the headline 1×10^{-3}. All headline Mamba runs completed under the frozen training protocol.

Architecture Summaries

Table 5 summarizes model size, representation dimension, pooling, and the main block structure used for each deep architecture. The detailed notes below the table spell out the input and classifier path without relying on long inline operator chains.

2D-CNN. Input tensors have shape $(1, 128, 1292)$. Each convolutional block applies convolution, batch normalization, ReLU, and time-frequency max pooling. Adaptive average pooling reduces the final feature map to $(1, 1)$ before a Linear(512, 16) classifier.

BiLSTM. The BiLSTM reads the mel sequence with two bidirectional recurrent layers, concatenating the two 256-dimensional directions into a 512-dimensional state. Attention-weighted temporal pooling produces the penultimate vector before a Linear(512, 16) classifier.

Transformer. Each time frame is projected from 128 mel bins to 256 dimensions and combined with sinusoidal positional encodings. The encoder uses four layers, four heads, $d_{\mathrm{ff}}{=}1024$, dropout 0.1, and a [CLS] token feeding a Linear(256, 16) classifier.

Mamba variants. Mamba-1 and Mamba-2 use the same input wrapper and classifier shape: five stacked sequence blocks, mean pooling over time, and a Linear(256, 16) classifier. They differ in the block implementation and state dimension: Mamba-1 uses $d_{\mathrm{state}}{=}16$, while Mamba-2 uses $d_{\mathrm{state}}{=}64$.

Feature Pipeline and 8-Group CKA Taxonomy

The classical feature matrix (540 dims) combines FMA's pre-computed 518-d DSP feature table with a 22-d project-defined rhythm vector. Eleven FMA feature families are used: MFCC (140); three chroma variants (84 each); spectral contrast (49); tonnetz (42); and five 7-d spectral or energy summaries. The added rhythm vector covers tempo, onset-envelope statistics, inter-beat intervals, and tempogram-energy statistics, using standard librosa extractors (McFee et al. 2015). This makes the classical baseline stronger than the raw FMA feature table alone and gives the representation analysis an explicit rhythm group, consistent with the timbre–pitch–rhythm framing of classic MIR features (Tzanetakis and Cook 2002). CKA decomposes the matrix into eight feature groups (372-d total): timbre (140), pitch class (84), tonal geometry (42), spectral contrast (49), spectral shape (21), noisiness (7), energy (7), and rhythm (22). Duplicate chroma variants are dropped from the CKA group selector to avoid triple-counting pitch-class energy but remain in the 540-d classical input matrix consumed by SVM, RF, and XGBoost.

Class Distribution and Trivial Baselines

The 16-genre distribution is severely imbalanced. The two largest training classes (Rock and Electronic) together account for roughly two-thirds of training examples; Easy Listening, International, and Blues each represent a fraction of a percent. On the benchmark-ready training split, the largest:smallest ratio is 4,878 Rock tracks to 13 Easy Listening tracks, or 375.2:1; over all usable tracks the corresponding corpus ratio is 338.8:1. Trivial baselines depend on the evaluation surface: a uniform-random 16-way prediction has 0.0625 expected accuracy, while an always-Rock predictor attains 0.3442 accuracy on the test split. Test supports for the long-tail classes are: International 2, Easy Listening 6, Blues 8, Spoken 12, Country 18, Pop 19, Jazz 39, Soul-RnB 42, Old-Time/Historic 51, Folk 52, Classical 62, Instrumental 74, Hip-Hop 120, Experimental 125, Electronic 532, and Rock 610.

Full Per-Class F1 Across All Eight Models

Values are seed means from the headline evaluation. Easy Listening and International are zero-F1 across all eight models; Pop, Blues, and (for several models) Country and Soul-RnB are near-zero. Random Forest reaches zero F1 on five long-tail classes despite class-balanced sample weights, illustrating that imbalance is a metric-level rather than only a model-level limitation.

Support-bin class-imbalance reanalysis.

Test support bin	Classes	Mean F1
≤ 10	Intl., Easy, Blues	0.000
11–25	Spoken, Country, Pop	0.178
26–75	six mid-support classes	0.371
76–200	Hip-Hop, Experimental	0.423
> 200	Electronic, Rock	0.682

This support-bin summary shows the same limitation from the Tier-1 robustness surface: classes with at most ten test examples have mean F1 of zero, while the two majority classes average 0.682, reinforcing that label support is a central ceiling on macro-F1.

Full Top-5 Confusion Table With Per-Group Distances

For each deep architecture, the top-5 confused (true, predicted) pairs by normalized off-diagonal mass, with the genre-centroid Euclidean distance computed independently in each of the eight CKA feature groups. "Sum" is normalized off-diagonal mass; full distance is the Euclidean distance over the 540-d classical matrix.

Rock $\leftrightarrow$ Soul-RnB appears in 5/5 architectures' top-5 lists; International-anchored pairs (test support 2) and Easy-Listening-anchored pairs (support 6) are partly driven by small support. The main table reports the full feature distance and rhythm distance; the omitted feature-group distances follow the same centroid-distance computation.

Full Pairwise Statistical Tests

The ten paired tests across the five deep architectures (paired t-test on macro-F1 differences across seeds 42–46; Bonferroni p uses family-wise α=0.05/10=0.005).

Zero pairs survive the Bonferroni correction. The three near-misses cluster around raw $p \approx 0.06$–0.09 with $|d|$ in the range 1.02–1.14, large-but-underpowered effects.

Multi-Init Untrained CKA Floor and Trained-vs-Untrained

Per-pair multi-init untrained linear CKA (mean $\pm$ std over five initialization seeds per architecture, ten cross-architecture pairs):

Pair	Untrained mean	std
2D-CNN $\leftrightarrow$ BiLSTM	0.305	0.026
2D-CNN $\leftrightarrow$ Transformer	0.327	0.045
2D-CNN $\leftrightarrow$ Mamba-1	0.353	0.038
2D-CNN $\leftrightarrow$ Mamba-2	0.341	0.076
BiLSTM $\leftrightarrow$ Transformer	0.922	0.005
BiLSTM $\leftrightarrow$ Mamba-1	0.938	0.003
BiLSTM $\leftrightarrow$ Mamba-2	0.840	0.021
Transformer $\leftrightarrow$ Mamba-1	0.934	0.011
Transformer $\leftrightarrow$ Mamba-2	0.855	0.020
Mamba-1 $\leftrightarrow$ Mamba-2	0.917	0.015
all 50 pair/init values	0.673	0.285

The floor is sharply bimodal: pairs that include 2D-CNN sit at 0.30–0.35 even at random initialization; pairs among the four sequence-style architectures already reach 0.84–0.94 untrained. The aggregate mean of 0.673 is a literal average of two regimes, not a uniform null.

As a kernel robustness check, the trained inter-architecture off-diagonal mean is 0.6853 with linear CKA and 0.6990 with RBF CKA on the same seed-42 representations, so the qualitative CKA story is not only a linear-kernel artifact.

Same-architecture trained-vs-untrained linear CKA, multi-init average over five init seeds:

Arch.	trained-vs-untrained mean	std
2D-CNN	0.269	0.010
BiLSTM	0.551	0.007
Transformer	0.396	0.034
Mamba-1	0.480	0.010
Mamba-2	0.475	0.031

2D-CNN's trained representation is by far the most distant from its untrained counterpart, consistent with its highest within-architecture cross-seed ceiling (0.877) and its strongest convolutional inductive bias being "burned in" during training.

Per-Genre Inter-Architecture CKA (Seed 42)

Off-diagonal mean linear CKA computed within each true-genre subset of the seed-42 representations. Smaller test supports (International n=2, Easy Listening n=6, Blues n=8, Spoken n=12) yield unstable CKA estimates—International produces an exact 1.0 by construction with only two samples—so the inverse correlation between sample size and per-genre CKA is partly a sample-size effect. We document the table for completeness, not as evidence that rare classes have higher learned similarity.

Genre	n	off-diag mean CKA
International	2	1.000
Easy Listening	6	0.826
Blues	8	0.728
Pop	19	0.701
Soul-RnB	42	0.678
Spoken	12	0.658
Instrumental	74	0.631
Hip-Hop	120	0.615
Country	18	0.614
Old-Time / Historic	51	0.609
Jazz	39	0.589
Folk	52	0.576
Experimental	125	0.564
Electronic	532	0.545
Classical	62	0.515
Rock	610	0.491

These values are computed from the seed-42 penultimate representations.

Full Pairwise Linear-Probe Transfer Matrix

Self-probe macro-F1 per architecture (diagonal): 2D-CNN 0.690, BiLSTM 0.573, Transformer 0.419, Mamba-1 0.487, Mamba-2 0.507. Off-diagonal source-probe-on-target via ridge linear bridge (rows are source, columns are target):

src \ tgt	2D-CNN	BiLSTM	Trans.	M-1	M-2
2D-CNN	0.690	0.464	0.406	0.431	0.527
BiLSTM	0.562	0.573	0.472	0.513	0.525
Transformer	0.548	0.467	0.419	0.455	0.481
Mamba-1	0.560	0.452	0.426	0.487	0.499
Mamba-2	0.559	0.491	0.461	0.555	0.507

Aggregate statistics: mean self-probe macro-F1 $= 0.535$; mean off-diagonal $= 0.493$; transfer retention $= 0.920$. Best off-diagonal: BiLSTM $\to$ 2D-CNN (macro-F1 0.562). Worst: 2D-CNN $\to$ Transformer (macro-F1 0.406).

Full Crop Ablation (15 Rows)

Each row is one completed 15-second-crop run. The 2D-CNN, BiLSTM, and Mamba-1 rows use FMA Full 60s tensors center-cropped to 15s; the Mamba-2 and Transformer rows use FMA Medium 30s tensors center-cropped to 15s. The Mamba-1 crop rows are post-retry runs at lr 2×10^{-4} rather than the headline lr 1×10^{-3}.

Arch.	Input	Seed	Macro-F1	Acc.
2D-CNN	full60	42	0.322	0.545
2D-CNN	full60	43	0.346	0.573
2D-CNN	full60	44	0.319	0.497
BiLSTM	full60	42	0.291	0.520
BiLSTM	full60	43	0.293	0.591
BiLSTM	full60	44	0.280	0.520
Mamba-1	full60	42	0.304	0.541
Mamba-1	full60	43	0.299	0.569
Mamba-1	full60	44	0.288	0.530
Mamba-2	medium	42	0.346	0.582
Mamba-2	medium	43	0.361	0.635
Mamba-2	medium	44	0.344	0.588
Transformer	medium	42	0.311	0.528
Transformer	medium	43	0.313	0.515
Transformer	medium	44	0.321	0.595

Mean by architecture: 2D-CNN 0.329, BiLSTM 0.288, Mamba-1 0.297, Mamba-2 0.351, Transformer 0.315.

Full SpecAugment Ablation

Macro-F1 with SpecAugment on (frozen seed-42 baseline) versus off (seed 42 only, all five architectures, 30-second mel input).

Arch.	SpecAug on	SpecAug off	Δ
2D-CNN	0.378	0.267	-0.111
Transformer	0.339	0.289	-0.050
Mamba-1	0.343	0.334	-0.010
Mamba-2	0.326	0.316	-0.010
BiLSTM	0.307	0.322	$+0.015$

Single-seed deltas; the BiLSTM gain is reported as an observed sensitivity result rather than a general architecture claim. The Mamba-1 no-SpecAugment run is confounded because it also used a lower learning rate (0.0002 rather than 0.001) and patience 8 rather than 10, so that row should not be interpreted as an augmentation-only estimate.

Full 60-Second Long-Sequence Pilot

Arch.	Seed	Macro-F1	Acc.
Mamba-2	42	0.317	0.512
Mamba-2	43	0.321	0.580
Mamba-2	44	0.354	0.647
Mamba-2	45	0.325	0.547
Mamba-2	46	0.336	0.535
Transformer	42	0.340	0.585
Transformer	43	0.326	0.562
Transformer	44	0.319	0.564

Mean macro-F1 by architecture: Mamba-2 0.331 (n=5, std 0.014), Transformer 0.328 (n=3, std 0.011). Frozen 30-second baselines for the same architectures: Mamba-2 0.356 ($\Delta = -0.025$), Transformer 0.316 ($\Delta = +0.012$). The comparison should be read as a context-length robustness check because the 60-second setting uses a distinct preprocessing surface.

Learning Curves (Single Seed)

Arch. / fraction	25%	50%	75%	100%
2D-CNN	0.354	0.305	0.329	0.340
BiLSTM	0.283	0.332	0.337	0.307
Transformer	0.306	0.254	0.277	0.311
Mamba-1	0.322	0.364	0.337	0.334
Mamba-2	0.339	0.327	0.332	0.340

Curves are seed-42 only and non-monotonic; we use them as diagnostics rather than as scaling laws. The corresponding plot is Figure 4.

Calibration: Extended Table

Arch.	Mean conf.	Mean entropy	Top-3
2D-CNN	0.565 ± 0.039	1.329 ± 0.134	0.802 ± 0.027
BiLSTM	0.664 ± 0.101	1.002 ± 0.334	0.816 ± 0.017
Transformer	0.548 ± 0.072	1.366 ± 0.242	0.806 ± 0.021
Mamba-1	0.658 ± 0.141	1.041 ± 0.446	0.814 ± 0.037
Mamba-2	0.629 ± 0.099	1.125 ± 0.318	0.816 ± 0.027

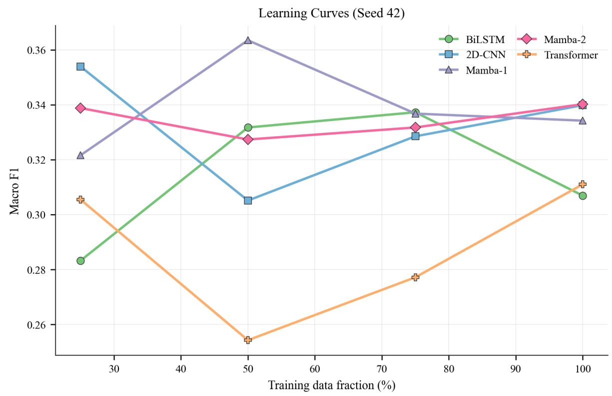

Figure 4: Per-architecture macro-F1 versus training-set fraction (seed 42 only). Non-monotonicity reflects single-seed noise and rare-class amplification rather than data inefficiency.

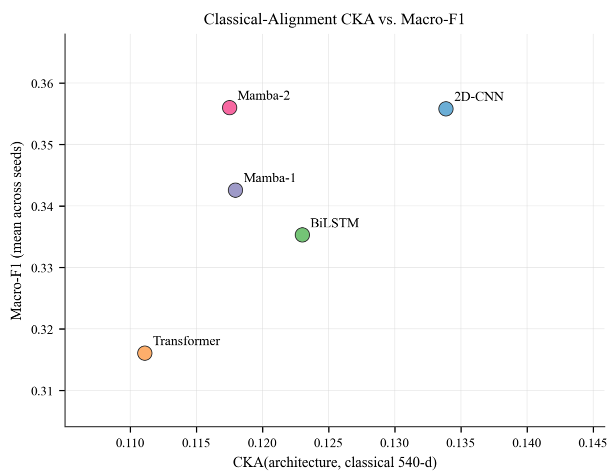

Figure 5: Per-architecture CKA against the full 540-d classical feature matrix versus mean macro-F1, with n=5 deep models (visual diagnostic only).

Reading: BiLSTM and Mamba-1 are the most confident (mean confidence ≈ 0.66) and lowest-entropy; Transformer is the least confident and most entropic, consistent with its weakest macro-F1. Top-3 accuracies cluster between 0.80 and 0.82 across all five architectures.

CKA-vs-F1 Scatter (Visual Diagnostic)

Figure 5 plots per-architecture mean CKA against the full 540-d classical feature matrix versus per-architecture mean macro-F1. With only five deep-model points, no Spearman or Pearson statistic is reported.

Other Visual Diagnostics: t-SNE, Time-of-Context, Per-Genre CKA

Three visual diagnostics supplement the main figures. (i) Seed-42 t-SNE projections of penultimate vectors per architecture use PCA to 50 dimensions and perplexity 30. (ii) Time-of-context curves at 60-second input cover Transformer seeds 42–44 (CLS-to-time attention) and Mamba-2 seed 46 (an activation/state-magnitude proxy, *not* a literal hidden SSM state). (iii) The per-genre inter-architecture CKA heatmap is dominated by sample-size effects and is included as appendix-only with that caveat.

H3 Detail: Per-Architecture Spearman Tests

Per-architecture Spearman ρ between the architecture's 8-vector of feature-group CKA scores and its mean per-group genre-centroid distance over the architecture's top-5 confused pairs ($n = 8$ groups; uncorrected p):

Arch.	ρ	p	sig.?
2D-CNN	+0.619	0.102	no
BiLSTM	+0.762	**0.028**	yes
Transformer	+0.667	0.071	no
Mamba-1	+0.714	**0.047**	yes
Mamba-2	+0.714	**0.047**	yes

All five ρ are positive; three of five reach uncorrected $p < 0.05$. None survive a conservative Bonferroni correction at α=0.05/5=0.01. We report this as directional support for H3, not as a corrected-significance result.

Reproducibility Notes

The frozen 40-run evaluation set is the canonical source for all main-paper numbers. The same admitted-run set defines headline aggregate metrics, pairwise tests, per-class metrics, confusion matrices, and CKA inputs. The frozen evaluation manifest is tracked by SHA-256 prefix `9a619...40f5ce`. This is included only as a provenance checksum for artifact matching.

CKA outputs. The representation analysis includes trained 5×5 inter-architecture matrices, the single-init untrained floor, within-architecture cross-seed ceiling, full and 8-group classical CKA, novelty gap, feature collinearity, and the multi-init untrained-floor comparison. Linear-probe transfer is reported as a supervised complement to these similarity scores.

Robustness outputs. The robustness suite includes the 15-second crop ablation, 60-second long-sequence pilot, SpecAugment-off ablation, reliability diagrams, class-imbalance reanalysis, and time-of-context analysis. These results are reported separately from the headline benchmark because they intentionally vary the input setting or augmentation setting.

Artifact scope. This repository copy contains paper-side and metric-side artifacts: manifests, aggregate metrics, per-class tables, CKA outputs, reliability files, figure assets, and source code. Some rerun inputs are pointer-only rather than copied locally: classical feature matrices and heavy checkpoint/array contracts for selected May-9 runs were pointer-validated on the WinPC location documented in the evidence index and large-artifact manifests.

Reproducibility protocol. The benchmark fixes dataset split, preprocessing, model capacity band, optimizer family, seed set, and metric aggregation before final comparison. Robustness analyses are identified by the specific experimental factor they vary.

Checklist	
Checklist item	Paper record
Dataset/splits	FMA Medium benchmark-ready split: 13,512 / 1,704 / 1,772, zero artist overlap.
Preprocessing	30-second log1p-compressed mel power spectrograms; 540-d classical matrix for SVM/RF/XGBoost.
Models	Eight reported models with parameter counts and architecture summaries in Appendix .
Seeds	Deep models use seeds 42–46; SVM-RBF is deterministic and reported with zero seed variance.
Training	AdamW, class weights, SpecAugment, epoch/patience schedule, and LR/dropout values are listed above.
Metrics	Macro-F1 primary; accuracy, weighted F1, balanced accuracy, calibration, and pairwise tests reported.
Analysis tools	Linear/RBF CKA, untrained floors, within-architecture ceilings, feature-group CKA, confusion analysis, linear-probe transfer.
Artifacts	Metric-side outputs are local; heavy checkpoints/arrays and classical matrices are documented as pointer-only where not copied.
Limits	No pretrained encoders, no SVM feature-group dropout, and no causal interpretation of CKA.

Table 2: Calibration and top-k accuracy for the five deep models (mean ± std). Lower is better for ECE and Brier.

Model	Top-2	ECE	Brier
2D-CNN	0.720 ± 0.036	**0.044 ± 0.013**	0.580 ± 0.053
BiLSTM	0.741 ± 0.027	0.094 ± 0.055	0.586 ± 0.046
Transformer	0.726 ± 0.018	0.075 ± 0.034	0.611 ± 0.032
Mamba-1	**0.749 ± 0.042**	0.095 ± 0.059	0.586 ± 0.079
Mamba-2	0.745 ± 0.032	0.083 ± 0.046	**0.567 ± 0.055**

Table 3: Per-architecture penultimate-layer CKA summary (seed-42 representations on the 1,772-track test set). The trained inter-architecture mean is high but lifts only marginally above the multi-init untrained floor; novelty gaps versus the 540-d classical feature matrix are uniformly large.

Arch.	inter	class.	gap	ceil.	untr.
2D-CNN	0.662	0.134	+0.528	**0.877**	0.255
BiLSTM	0.680	0.123	+0.557	0.830	0.544
Transformer	0.622	0.111	+0.511	0.748	0.444
Mamba-1	0.730	0.118	+0.612	0.750	0.484
Mamba-2	**0.732**	0.118	**+0.615**	0.790	0.479

Trained off-diag mean = 0.685; multi-init untrained floor = 0.673; Within-arch ceiling mean = 0.799; trained−floor = 0.012.

Table 4: Top-3 confused pairs per deep architecture by normalized off-diagonal mass (seed-aggregated). Rock ↔ Soul-RnB appears in 5/5 architectures' top-5 lists.

Architecture	Pair	Mass
2D-CNN	Easy Listening ↔ Instrumental	0.541
	Rock ↔ Soul-RnB	0.508
	International ↔ Jazz	0.500
BiLSTM	International ↔ Rock	0.600
	Folk ↔ Instrumental	0.413
	Electronic ↔ Pop	0.394
Transformer	International ↔ Jazz	**0.815**
	Easy Listening ↔ Instrumental	0.433
	Country ↔ Rock	0.426
Mamba-1	International ↔ Rock	0.501
	Electronic ↔ Pop	0.439
	Blues ↔ Rock	0.404
Mamba-2	Classical ↔ Easy Listening	0.500
	International ↔ Jazz	0.500
	Rock ↔ Soul-RnB	0.405

Table 5: Deep architecture summaries. "Pen." is the penultimate representation dimension used for CKA and linear-probe transfer.

Model	Params	Pen.	Pooling	Main structure
2D-CNN	4,105,552	512	Adaptive avg. pool	Five convolutional blocks over the log-mel image; channel widths 1, 32, 128, 256, 512, 512.
BiLSTM	2,442,513	512	Attention pool	Two bidirectional LSTM layers with hidden size 256 per direction.
Transformer	3,196,944	256	[CLS] token	Linear projection from 128 mel bins to $d_{\mathrm{model}}=256$, sinusoidal positions, and four encoder blocks.
Mamba-1	2,229,008	256	Temporal mean	Five Mamba blocks with $d_{\mathrm{model}}=256$, $d_{\mathrm{state}}=16$, $d_{\mathrm{conv}}=4$, expand 2.
Mamba-2	2,199,048	256	Temporal mean	Same wrapper as Mamba-1, using Mamba-2 blocks with $d_{\mathrm{state}}=64$.

Table 6: Complete classical feature table used by SVM-RBF, Random Forest, and XGBoost. Source citations: FMA features (Defferrard et al. 2017); rhythm extraction (McFee et al. 2015); descriptor anchors (Tzanetakis and Cook 2002; Logan 2000; Bartsch and Wakefield 2005; Müller and Ewert 2011; Jiang et al. 2002; Harte, Sandler, and Gasser 2006; Peeters 2004). FMA families contribute seven statistics per coefficient; CKA uses the listed 372-d subset.

Source	Family / subgroup	Dim.	Descriptor	CKA group
FMA	MFCC	140	20 cepstral coefficients $\times$ 7 statistics; timbral envelope	timbre
FMA	chroma CENS	84	12 energy-normalized pitch-class coefficients $\times$ 7 statistics	pitch class
FMA	chroma CQT	84	12 constant-Q pitch-class coefficients $\times$ 7 statistics	included in 540-d matrix only
FMA	chroma STFT	84	12 STFT pitch-class coefficients $\times$ 7 statistics	included in 540-d matrix only
FMA	spectral contrast	49	7 sub-band contrast coefficients $\times$ 7 statistics	spectral contrast
FMA	tonnetz	42	6 tonal-centroid coordinates $\times$ 7 statistics	tonal geometry
FMA	RMS energy	7	1 energy coefficient $\times$ 7 statistics	energy
FMA	spectral centroid	7	1 brightness-centroid coefficient $\times$ 7 statistics	spectral shape
FMA	spectral bandwidth	7	1 bandwidth coefficient $\times$ 7 statistics	spectral shape
FMA	spectral rolloff	7	1 rolloff coefficient $\times$ 7 statistics	spectral shape
FMA	zero-crossing rate	7	1 zero-crossing coefficient $\times$ 7 statistics; noisiness proxy	noisiness
Added rhythm	global tempo	1	Beat-tracking BPM scalar	rhythm
Added rhythm	onset envelope	7	Summary statistics of onset strength	rhythm
Added rhythm	inter-beat intervals	7	Summary statistics of adjacent beat intervals in seconds	rhythm
Added rhythm	tempogram energy	7	Summary statistics of tempogram-energy structure	rhythm
Classical input	all feature rows	540	Full matrix consumed by SVM/RF/XGBoost	full baseline input
CKA grouped subset	selected groups	372	Duplicate chroma variants removed for grouped CKA	eight CKA groups

Table 7: Full per-class F1 across all eight models.

Genre	2D-CNN	BiLSTM	Trans.	M-1	M-2	SVM	RF	XGB
Blues	0.090	0.000	0.020	0.000	0.000	0.000	0.000	0.000
Classical	0.653	0.641	0.592	0.692	0.674	0.821	0.837	0.564
Country	0.145	0.105	0.067	0.200	0.178	0.071	0.000	0.198
Easy Listening	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Electronic	0.649	0.665	0.648	0.692	0.715	0.801	0.724	0.703
Experimental	0.319	0.262	0.194	0.283	0.320	0.430	0.159	0.351
Folk	0.393	0.321	0.325	0.369	0.368	0.444	0.449	0.508
Hip-Hop	0.643	0.610	0.548	0.587	0.665	0.690	0.550	0.605
Instrumental	0.173	0.153	0.114	0.128	0.119	0.104	0.000	0.167
International	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Jazz	0.342	0.368	0.342	0.341	0.362	0.619	0.177	0.435
Old-Time	0.926	0.939	0.863	0.932	0.942	0.971	0.978	0.923
Pop	0.019	0.013	0.013	0.015	0.014	0.000	0.000	0.048
Rock	0.744	0.760	0.731	0.727	0.745	0.853	0.804	0.817
Soul-RnB	0.152	0.136	0.063	0.079	0.148	0.040	0.000	0.138
Spoken	0.445	0.392	0.538	0.438	0.446	0.545	0.483	0.507

Table 8: Full top-5 confusion table with selected per-group distances.

Arch.	Pair	Sum	d_{full}	d_{rhythm}
2D-CNN	EasyL. $\leftrightarrow$ Instr.	0.541	12.18	4.62
	Rock $\leftrightarrow$ Soul-RnB	0.508	10.70	4.67
	Intl. $\leftrightarrow$ Jazz	0.500	10.26	1.45
	Folk $\leftrightarrow$ Instr.	0.435	7.66	2.84
	Class. $\leftrightarrow$ EasyL.	0.433	15.95	3.84
BiLSTM	Intl. $\leftrightarrow$ Rock	0.600	12.74	2.17
	Folk $\leftrightarrow$ Instr.	0.413	7.66	2.84
	Electr. $\leftrightarrow$ Pop	0.394	7.03	1.79
	Rock $\leftrightarrow$ Soul-RnB	0.377	10.70	4.67
	Class. $\leftrightarrow$ EasyL.	0.367	15.95	3.84
Trans.	Intl. $\leftrightarrow$ Jazz	**0.815**	10.26	1.45
	EasyL. $\leftrightarrow$ Instr.	0.433	12.18	4.62
	Country $\leftrightarrow$ Rock	0.426	5.71	1.58
	Rock $\leftrightarrow$ Soul-RnB	0.409	10.70	4.67
	Blues $\leftrightarrow$ Rock	0.404	8.75	2.29
M-1	Intl. $\leftrightarrow$ Rock	0.501	12.74	2.17
	Electr. $\leftrightarrow$ Pop	0.439	7.03	1.79
	Blues $\leftrightarrow$ Rock	0.404	8.75	2.29
	Folk $\leftrightarrow$ Instr.	0.358	7.66	2.84
	Rock $\leftrightarrow$ Soul-RnB	0.346	10.70	4.67
M-2	Class. $\leftrightarrow$ EasyL.	0.500	15.95	3.84
	Intl. $\leftrightarrow$ Jazz	0.500	10.26	1.45
	Rock $\leftrightarrow$ Soul-RnB	0.405	10.70	4.67
	Blues $\leftrightarrow$ Rock	0.402	8.75	2.29
	Intl. $\leftrightarrow$ Rock	0.400	12.74	2.17

Table 9: Full pairwise statistical tests across deep architectures.

Pair (A vs B)	Diff (A−B)	d	t	p_{raw}	p_{Bonf}
2D-CNN vs BiLSTM	$+0.020$	$+0.52$	1.17	0.307	1.000
2D-CNN vs Trans.	$+0.040$	$+1.14$	2.54	0.064	0.640
2D-CNN vs M-1	$+0.013$	$+0.27$	0.60	0.578	1.000
2D-CNN vs M-2	-0.000	-0.01	-0.01	0.990	1.000
BiLSTM vs Trans.	$+0.019$	$+0.59$	1.31	0.261	1.000
BiLSTM vs M-1	-0.007	-0.16	-0.35	0.747	1.000
BiLSTM vs M-2	-0.021	-1.05	-2.35	0.078	0.782
Trans. vs M-1	-0.027	-0.44	-0.98	0.383	1.000
Trans. vs M-2	-0.040	-1.02	-2.28	0.085	0.846
M-1 vs M-2	-0.013	-0.24	-0.53	0.621	1.000